END SEMESTER ASSESSMENT (ESA)

B.TECH. (CSE)

IV SEMESTER

UE21CS251B – MICROPROCESSOR AND COMPUTER ARCHITECTURE LABORATORY

PROJECT REPORT

ON

SMART IRRIGATION SYSTEM

SUBMITTED BY

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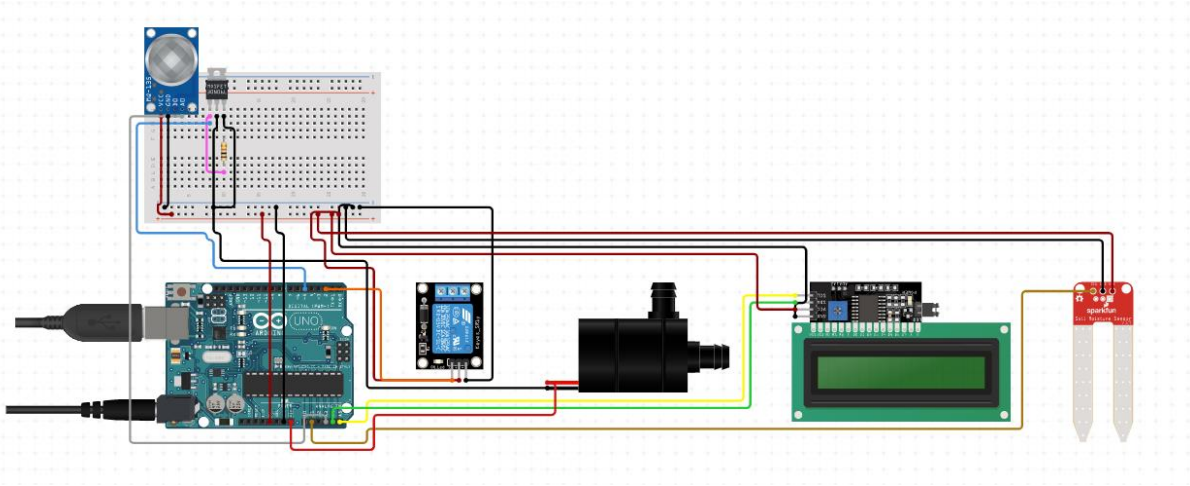
ABSTRACT OF THE PROJECT

Irrigation is the artificial application of water for the success of crop production in the field.
An automated irrigation system has been developed using sensors technology with Arduino to efficiently utilize water for irrigation purpose. The system has soil moisture sensor inserted into the soil of the plants and a water level sensor placed in a water container from where water will be pumped to plants for irrigation. A code in Arduino IDE has been build out to deal with threshold values of soil moisture and display the information in an LCD display .

Soil moisture sensor will detect dankness content of soil and accordingly turn ON/OFF the pumping motor. Human efforts can be reduced using this technique and increase saving of water by efficiently irrigating the plants.

Programmable micro-controller collects the input signals converted into values of moisture in the soil via soil moisture sensors. As the microcontroller starts obtaining the signals, it creates an output that forces a relay for running the water pumping motor. An LCD screen is also linked to the micro-controller to show moisture conditions of the soil .

CIRCUIT DIAGRAM



ARDUINO CODE

```
#include <LiquidCrystal_I2C.h>
#include <MQ135.h>

MQ135 gasSensor = MQ135(A0);
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  Serial.begin(9600);
  lcd.init();
  lcd.backlight();
  lcd.clear();
  pinMode(2, OUTPUT);
  digitalWrite(2, HIGH);
  delay(1000);

  // Print air quality data at the start
  int airQuality = gasSensor.getPPM();
  lcd.setCursor(0, 0);
  lcd.print("Air Quality: ");
  if (airQuality < 200) {
    lcd.setCursor(0, 1);
    lcd.print("Good");
  } else if (airQuality < 400) {
    lcd.setCursor(0, 1);
    lcd.print("Medium");
  } else {
    lcd.setCursor(0, 1);
    lcd.print("Bad");
  }
  delay(6000);
  lcd.clear();

  lcd.setCursor(0, 0);
  lcd.print("IRRIGATION");
  lcd.setCursor(0, 1);
  lcd.print("SYSTEM IS ON ");
  lcd.print("");
  delay(3000);
  lcd.clear();
```

```
}

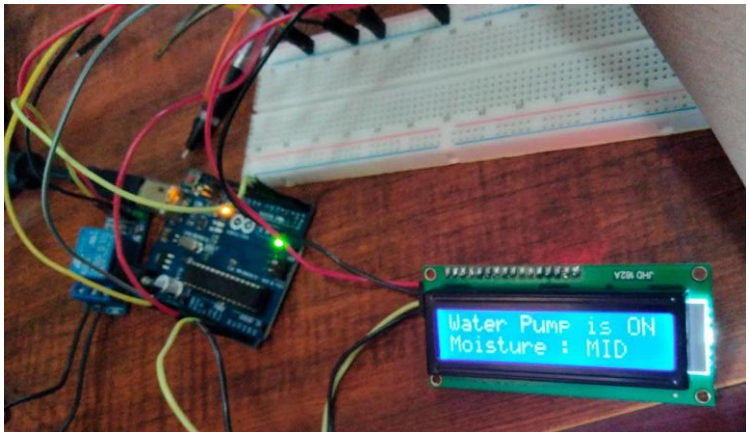
void loop() {
  int moisture = analogRead(A1);
  int airQuality = gasSensor.getPPM();

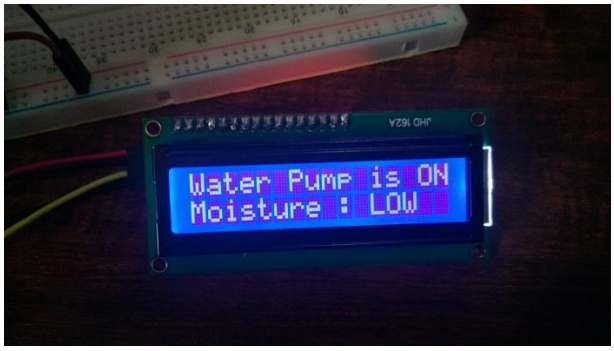
  // Control water pump based on moisture level
  if (moisture > 1200) {
    digitalWrite(2, LOW);
    lcd.setCursor(0, 0);
    lcd.print("Water Pump is ON ");
  } else {
    digitalWrite(2, HIGH);
    lcd.setCursor(0, 0);
    lcd.print("Water Pump is OFF");
  }

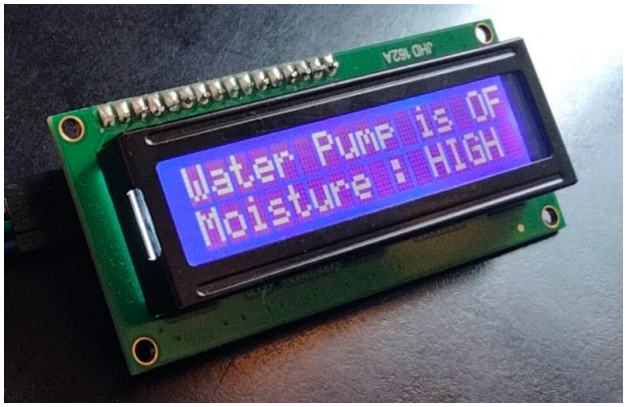
  // Display moisture level
  if (moisture < 300 ) {
    lcd.setCursor(0, 1);
    lcd.print("Moisture : HIGH");
  } else if (moisture > 300 && moisture < 1200) {
    lcd.setCursor(0, 1);
    lcd.print("Moisture : MID ");
  } else if (moisture > 1200 ) {
    lcd.setCursor(0, 1);
    lcd.print("Moisture : LOW ");
  }
}
```

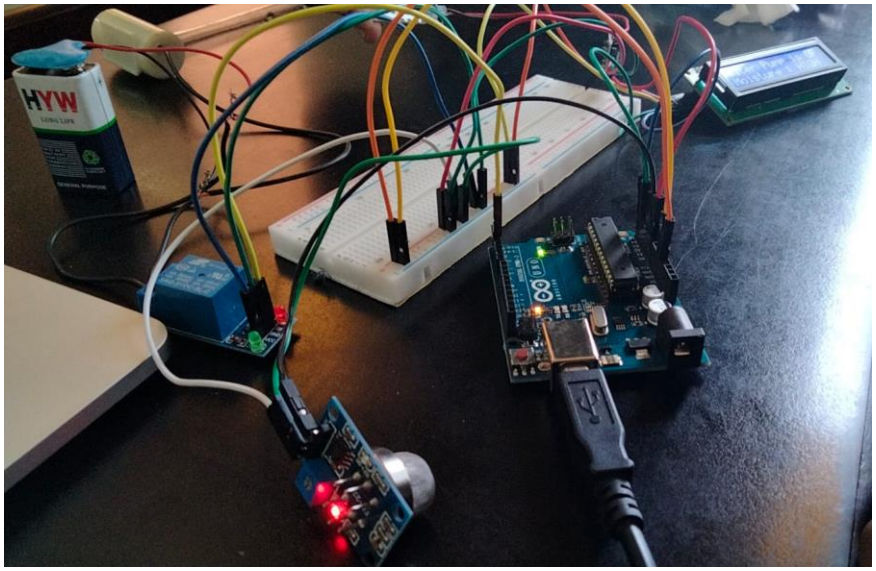
SCREEN SHOTS OF THE OUTPUT

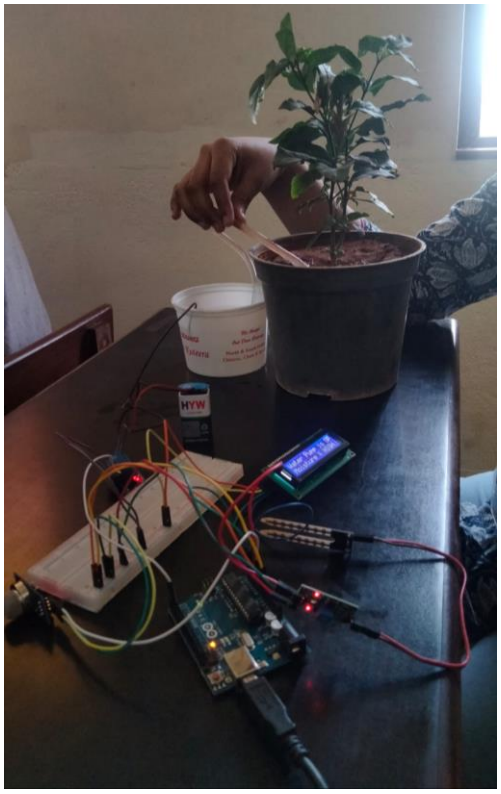


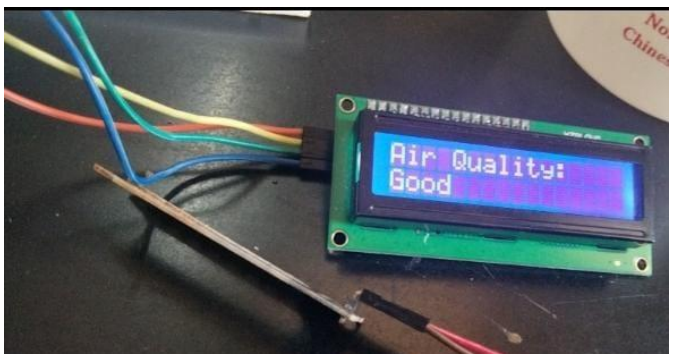







HYW


HYW


Air Quality:
Good

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